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Strength and Power Thresholds to Identify High and Low Linear Sprint Speed Performers in Collegiate American Football Players

Nicolas M. Philipp,[1] Derek A. Crawford,[2] Dimitrije Cabarkapa,[1] and Andrew C. Fry[1]

[1]*Jayhawk Athletic Performance Laboratory, Wu Tsai Human Performance Alliance, University of Kansas, University of Kansas, Lawrence, Kansas; and* [2]*Department of Nutrition, Kinesiology, and Health, University of Central Missouri, Warrensburg, Missouri*

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Abstract

Philipp, NM, Crawford, DA, Cabarkapa, D, and Fry, AC. Strength and power thresholds to identify high and low linear sprint speed performers in collegiate American football players. *J Strength Cond Res* XX(X): 000–000, 2023—Lower-body strength and power are commonly measured performance qualities across a number of sports. In recent years, more attention has been given to relationships, primarily between lower-body strength and linear speed performance. While still limited, evidence is in agreement that lower-body strength positively contributes to linear speed performance. However, what is less well understood is if there comes a point in an athlete's development, at which, further working on increasing maximal strength may not fully compliment additional gains in speed performance. Within this study, authors aimed to provide practitioners with lower-body strength and power thresholds that can discriminate between slow and fast performers, within a group of collegiate American football players. The sample was further divided into a high-body and low-body weight group, and authors hypothesized that by using logistic regression, supplemented with receiver operator curve analyses, optimal cut-off points (i.e., relative lower-body strength thresholds) that are able to significantly discriminate between slow and fast linear speed performers may be identified. Findings indicate that optimal cut-off scores differed between the groups of athletes, as well as the lower body strength and power tests. All models were able to significantly distinguish between slower and faster performers, and area under the curve values ranged from 0.695 to 0.903. Although thresholds will likely vary based on factors such as sex, training age, and sport, findings from this investigation may be used to benchmark athletes and to further individualize training aimed at improving linear speed performance.

Key Words: sport, sport science, college football, sprinting

Introduction

Lower-body strength and power have been shown to be one of the driving factors of sport success from a physical standpoint (2–5,7,16,20,23,33). Recent investigations have shown correlations between absolute and relative lower-body strength and performance within different athletic key performance indicators such as jumping, sprinting, and changing directions efficiently (2,21,22,24,30). More specifically, the relationship between lower-body strength and linear sprint performance has been of interest to practitioners and researchers within previously conducted studies (5,20,21,24,27,29). A growing body of literature suggests substantial relationships between strength and sprinting performance, with increases in strength coinciding with improvements in sprint performance over shorter distances (27). For instance, McBride et al. (20) showed that 1 repetition maximal strength, divided by body mass within the back squat exercise, was significantly related to sprinting ability within male athletes. Furthermore, Wisloff et al. (32) reported a strong, significant correlation between 1 repetition maximum (1RM) strength within the squat exercise and 10-meter sprint times. More recently, in a meta-analysis, Seitz et al. (27) suggested that resistance training induced increases in lower-body strength transfer positively to improvements in sprint performance. However, this meta-analysis received some criticism regarding potential statistical misinterpretations, making it difficult to draw solid conclusions from this study (9). In 2012, Comfort et al. (5) compared a sample of professional Rugby athletes to a group of recreationally trained athletes regarding relative lower-body strength and 5-m, 10-m, and 20-m sprint performance. In this study, it was suggested that relative lower-body strength is of importance for initial sprint acceleration in all athletes but was more strongly correlated to sprint performance over greater distances in the group of recreationally trained athletes who were lighter and weaker, looking at absolute strength levels. Furthermore, Philipp et al. (24) recently showed that collegiate American football players experienced significant increases in relative and absolute lower-body strength and total body mass over the course of 3 athletic seasons, which were not accompanied by significant improvements in linear speed performance. Interestingly, when the study sample was divided into a high and a low strength group, based on initial, season 1 relative strength levels, significant improvement in speed were observed for the low strength group, whereas the high strength group did not experience significant improvements in speed. The aforementioned findings raise the question of whether there is an optimal strength threshold above which further developing maximal strength fails to complement improvements in dynamic tasks such as sprinting.

Furthering this thought, a commonly discussed concept within the realms of strength and conditioning is the use of

Address correspondence to Nicolas M. Philipp, nicophilipp@ku.edu.

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thresholds to guide decisions on the implementation of training or other interventions. In the sports science literature, thresholds have been used in different ways. From a pathological standpoint for instance, interlimb-asymmetry thresholds have been used to make decisions for return to play scenarios after injury (14). Similarly, thresholds have been established to gauge potential injury risk based on certain scores on physical tests or assessments (11). When examining lower-body strength through a more performance-oriented lens, practitioners continue to discuss and show interest toward strength thresholds, or ratios, specifically aiming at the question of "how strong is strong enough." Stone (28), as well as Lucero et al. (16), has suggested that in sports such as power lifting or Olympic style weightlifting, continuous increases in strength over time would show themselves as advantageous. However, there is still significant debate about how strong athletes in other sports should be. Although answering this question is multifactorial, and not the primary aim of this investigation, our study seeks to provide practitioners with lower-body strength and power thresholds that can discriminate between slow and fast performers, specifically looking at linear sprint speed performance. Although proposed thresholds will likely be highly task-specific and population-specific, complimentary methodologies such as longitudinal research designs may be applied to gain actionable insights into the relationships between lower-body strength and power and linear speed over time. We hypothesize that by using logistic regression, supplemented with receiver operator curve (ROC) analyses, we can determine optimal cut-off points (i.e., relative lower-body strength thresholds) that are able to significantly discriminate between slow and fast linear speed performers. Such thresholds may help practitioners individualize their approach toward implementing training strategies, providing actionable targets for their athletes. Furthermore, results from this investigation may be used for benchmarking purposes within similar populations.

Methods

Experimental Approach to the Problem

This investigation used physical performance data from a National Collegiate Athletic Association (NCAA) Division 2 American Football team, for the athletic seasons of 2015, 2016, 2017, 2018, 2019, 2020, and 2021. An ex post facto quasiexperimental study design was used to examine lower-body strength associations with linear sprint speed in collegiate American football players of differing body mass. To account for the effect of body mass on relative strength levels and speed, the sample was further divided into a "high body mass" (n = 147, body mass = 116.4 ± 13.7 kg) and a "low body mass" (n = 147, body mass = 86.4 ± 6.1) group, using a median split analyses for body weight. For athletes who had observations over multiple seasons, the mean of each performance variable was used for further analyses. The physical performance data was provided by the head strength and conditioning coach at the respective NCAA-Division 2 university and consisted of off-season test results for on-field tests, resembling the National Football League combine, as well as training roombased strength measures, such as the barbell back squat, bench press, or Olympic weightlifting derivative exercises. Of specific interest were data from the 40-yard dash linear speed test, as well as barbell back squat estimated 1RM, hang clean exercise estimated 1RM, vertical jump, and broad jump. All respective tests were performed and measured annually, at the end of each respective spring off-season (middle of April), with the months of January through March consisting of 4 weekly strength training sessions and 2 weekly speed and change of direction sessions. Testing was performed in the early afternoon hours, and proper execution of tests was ensured through visual observation and verbal feedback by a group of certified strength and conditioning specialists. This study used an ex post facto quasiexperimental study design, using deidentified data, provided and approved by the head strength and conditioning coach at the respective NCAA-Division 2 school. Owing to the data being retrospective and deidentified, the respective Kansas University's Institutional Review Board approved use of these data.

Subjects

The sample for this investigation consisted of a total of 294 collegiate NCAA-Division 2 American football players (body mass = 100.98 ± 18.18 kg), between the ages of 18 and 23 years. To account for the effect of body mass on relative strength levels and speed, the sample was further divided into a "high body mass" (n = 147, body mass = 116.4 ± 13.7 kg) and a "low body mass" (n = 147, body mass = 86.4 ± 6.1) group, using a median split analyses for body mass. All athletes were cleared for full sport participation by the sports medicine staff at the point of data collection.

Testing Procedures

Body Mass. Body mass was measured using an electronic Befour PS 7700 scale (Befour, Saukville, WI) and recorded in kilograms (kg).

Forty Yard Dash. The 40-yd dash was conducted with subjects beginning in a 3-point stance and running as fast as possible for 40 yards. According to Mann et al. (18), hand timing allows for an accurate classification of 40-yard dash performance among college football players. Lanes on the track gave guidance to the subjects. The best of 3 attempts was used in analysis. Timers started their handheld watch on the first movement of the athlete and stopped their watch when the athlete's upper body crossed the finish line (18). The same sport coach using the same stopwatch was used for all trials during all testing years.

Barbell Back Squat. The barbell back squats were performed inside a standard squat rack, with spotters placed behind the athlete, as well as on both ends of the barbell for safety reasons (total of 3 spotters). Certified strength and conditioning specialists ensured all squats were performed correctly. For a repetition to count, an athlete's femur had to be parallel to the ground. Again, certified strength and conditioning specialists visually observed each repetition to ensure this criterion was being met. Athletes' 1 repetition maximum was calculated from a 3-repetition maximum test. The Epley equation was used to calculate a 1RM from a 3-repetition maximum test (6). It has been suggested that performing repetitions closer to the 1RM (e.g., less than 10 repetitions) presents with greater accuracy in estimating 1RM strength, compared with performing a greater number of repetitions (25). Relative squat strength was calculated as the ratio between the raw 1RM number in kilograms divided by the athletes' body mass in kilograms.

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Hang Clean. The hang clean exercise was performed in front of the standard squat rack, with no spotters around the athletes. Again, certified strength and conditioning specialists ensured that all repetitions were performed correctly. Exercise execution was performed using the hang clean technique as previously established (10). Testing started with the athletes holding on to the barbell then lowering the barbell to the midthigh area above the knee, followed by explosively lifting the barbell upward, and catching it on the anterior deltoid muscles in a quarter squat position. Similarly, according to Kawamori et al. (10), a repetition was not counted as successful if it was caught in a squat position with the upper thigh below parallel to the floor. Athletes were permitted to use weightlifting wrist straps to avoid grip strength being a limiting factor. The athletes' 1RM was calculated from a 3RM test. The Epley equation was used to calculate a 1RM from a 3RM test (6). Relative hang clean strength was calculated as the ratio between the raw 1RM number in kilograms, divided by the athletes' body mass in kilograms.

Broad Jump. For the broad jump, procedures were similar to previous research, suggesting good reliability and validity of this assessment (1,13). Athletes began from a standing position and jumped as far as possible horizontally. A countermovement was used, and jump distance was noted on a steel measuring tape. The distance from the starting line to the rearmost heel strike was marked and measured. Athletes were allowed 3 attempts with the best being used for analysis. A successful attempt involved a jump from the starting line and the subjects landing without falling backward. All broad jumps were recorded in meters.

Vertical Jump. The countermovement vertical jump data was recorded using the Just Jump System (Probotics, Inc., Huntsville, AL). This jump mat provides an estimate of jump height calculated from flight time and has been validated against a motion capture system in previous research (15). Subjects were instructed to step onto the mat, to drop forcefully to their preferred depth, and to jump as fast and as high as possible. The task was performed with the use of a bilateral arm swing. Subjects were allowed 3 attempts with the best attempt being used for analysis. A successful attempt involved the subjects jumping from, and landing on, the measuring mat, maintaining straight legs throughout the flight phase, not bending their legs before landing, and elongating the flight phase. All vertical jumps were recorded in centimeters.

Statistical Analyses

Linear speed was assessed by the 40-yard dash test, and athletes within respective subgroups were classified as "slower" or "faster" performers, using the 50th percentile as cut-off points (median-split analysis). Furthermore, we used logistic regression analyses, supplemented with the use of ROC for each lower-body relative strength and power test, within each respective subgroup of athletes, to establish cut-off scores (i.e., thresholds) that maximized the positive predictive value (i.e., sensitivity) and minimized the false positive rate (i.e., 1 specificity) associated with "slower" and "faster" performers. Therefore, the optimal cut-off point on the curve was identified when the true-positive rate (sensitivity) was maximized, and the false-positive rate (1-specificity) was minimized, identified using the Youden Index as part of the "ROCit" package within the RStudio Software (Version 1.4.1106). The same package was used to plot respective ROC curves and to identify model summary statistics of interest (i.e., accuracy, specificity, sensitivity, and area under the curve [AUC]). Statistical inferences were made using an alpha level of $p \leq 0.05$.

Table 1

Receiver operator curve summary statistics for all lower-body strength and power assessments within the "high body mass" group.*

Metric	Accuracy	Specificity	Sensitivity	AUC	Cut-off (threshold)
Relative squat (ratio)	0.735	0.743	0.726	0.780	2.01
Relative hang clean (ratio)	0.755	0.757	0.753	0.848	1.28
Broad jump (m)	0.777	0.759	0.795	0.883	2.64
Vertical jump (cm)	0.828	0.810	0.846	0.903	72.64

*AUC = area under the curve; KS Cutoff = Kolmogorov-Smirnov cut-off.

Results

The ROC curve analyses revealed that relative lower-body strength scores for all tests, within the 2 body mass subgroups were able to significantly discriminate between "slow" and "fast" performers within the 40-yard dash linear speed test. However, different diagnostic utilities were observed between tests and subgroups and can be seen within Tables 1 and 2. Figure 1 highlights ROC curves for respective tests and subgroups.

Heavy Group

A significant ROC curve (AUC = 0.780, 95% CI = 0.705–0.855; $p = <0.001$) produced a relative lower-body strength threshold score (i.e., 2.01 1RM/BW) for the back squat exercise that was able to correctly classify 55 of 74 athletes (74.3%) who were slower performers, and 53 of 73 (72.6%) who were faster performers. Furthermore, a significant ROC curve (AUC = 0.848, 95% CI = 0.785–0.912; $p = <0.001$) produced a relative lower-body strength threshold score (i.e., 1.28 1RM/BW) for the hang clean exercise that was able to correctly classify 56 of 74 athletes (75.7%) who were slower performers and 55 of 74 (75.3%) who were faster performers. Looking at lower-body power, a significant ROC curve (AUC = 0.883, 95% CI = 0.829–0.937; $p = <0.001$) produced a lower-body power threshold score (i.e., 2.64 m) for the broad jump exercise that was able to correctly classify 60 of 79 athletes (75.9%) who were slower performers

Table 2

Receiver operator curve summary statistics for all lower-body strength and power assessments within the "low body mass" group.*

Metric	Accuracy	Specificity	Sensitivity	AUC	Cut-off (threshold)
Relative squat (ratio)	0.639	0.649	0.630	0.751	2.31
Relative hang clean (ratio)	0.612	0.622	0.603	0.695	1.56
Broad jump (m)	0.673	0.691	0.654	0.753	2.87
Vertical jump (cm)	0.704	0.741	0.667	0.779	84.1

*AUC = area under the curve; KS Cutoff = Kolmogorov-Smirnov cut-off.

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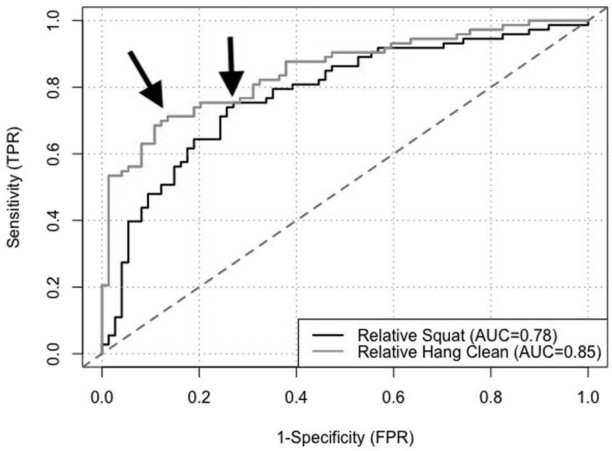

Figure 1. Receiver operator curves and AUC values for the "heavy" group and lower-body relative strength assessments. TPR = true positive rate; FPR = false positive rate; AUC = area under the curve. Arrows indicate location on curve where sensitivity is maximized and 1 specificity is minimized.

and 62 of 16 (79.5%) who were faster performers. Finally, a significant ROC curve (AUC = 0.903, 95% CI = 0.854–0.952; p = <0.001) produced a lower threshold score (i.e., 72.64 cm) for the vertical jump exercise that was able to correctly classify 64 of 79 athletes (81.0%) who were slower performers and 66 of 78 (84.6%) who were faster performers (Figure 2).

Light Group

A significant ROC curve (AUC = 0.751, 95% CI = 0.672–0.829; p = <0.001) produced a relative lower-body strength threshold score (i.e., 2.31 1RM/BW) for the back squat exercise that was able to correctly classify 48 of 74 athletes (64.9%) who were slower performers and 46 of 73 (63.0%) who were faster performers. Furthermore, a significant ROC curve (AUC = 0.695, 95% CI = 0.610–0.780; p = <0.001) produced a relative lower-body strength threshold score (i.e., 1.56 1RM/BW) for the hang clean exercise that was able to correctly classify 46 of 74 athletes (62.2%) who were slower performers and 44 of 73 (60.3%) who were faster performers. Looking at lower-body power, a significant ROC curve (AUC = 0.753, 95% CI = 0.677–0.829; p = <0.001) produced a lower-body power threshold score (i.e., 2.87 cm) for the broad jump exercise that was able to correctly classify 56 of 81 athletes (69.1%) who were slower performers and 51 of 78 (65.4%) who were faster performers. Finally, a significant ROC curve (AUC = 0.779, 95% CI = 0.706–0.851; p = <0.001) produced a lower-body power threshold score (i.e., 84.07 cm) for the vertical jump exercise that was able to correctly classify 60 of 81 (74.1%) athletes who were slower performers and 52 of 78 (66.7%) who were faster performers (Figures 3,4).

Discussion

The current study sought to determine whether meaningful thresholds of lower-body strength relative to body weight as well as lower-body power were capable of differentiating between slow and fast performers within the 40-yard dash linear sprint speed test. The associated purpose was to provide practitioners with potential benchmarks and actionable targets to help guide and individualize their training implementations, which may be used in further investigations by implementing multiyear research designs. A growing body of literature suggests substantial relationships between strength and sprinting performance, with increases in strength coinciding with improvements in sprint performance over shorter distances, in populations such as soccer and rugby (30). However, research reports have also suggested a common trend in which collegiate football players experience gains in strength and body mass over the course of their collegiate career, which are not accompanied by improvements in speed (8,19,24). For instance, Mann et al. (19) recently suggested that over the course of their 4-year collegiate career, American football players experienced significant increases in body mass, which were reflected in significant increases in sprint momentum but not sprint velocity. This makes it difficult to discern to what point improvements in strength aid in the development of speed and to what extent changes in anthropometric measures factor into this discussion.

Although the data used in this study were not longitudinal, findings from this investigation revealed that optimal cut-off scores differed between the 2 body massbased groups of athletes, as well as the lower-body strength and power tests, which may be used as training targets within the applied strength and conditioning setting. All models were able to significantly distinguish between slower and faster performers, and AUC values ranged from 0.695 to 0.903.

To the best of our knowledge, this is the first study within competitive sports, and particularly within collegiate American football that is investigating thresholds of relative lower-body strength and power, as they relate to linear sprint speed performance. As expected, due to their respective body masses, the high body mass group reported lower strength and power thresholds, whereas the low body mass group reported the larger strength and power thresholds. This simple finding alone should discourage practitioners from using arbitrary thresholds such as 2 times body mass to inform their decision making and planning of training implementations. Optimal thresholds are likely to be sex-specific, population-specific, and training agespecific, at a minimum.

The strongest model within our investigation was found when looking at the vertical jump test within the high body mass group. With an AUC of 0.903, and an optimal cut-off score of 72.64 cm,

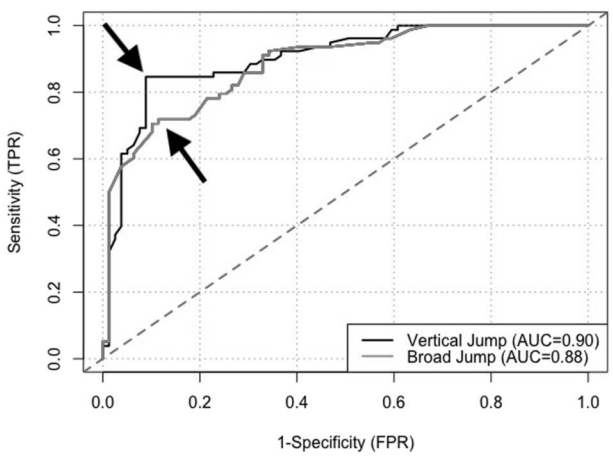

Figure 2. Receiver operator curves and AUC values for the "heavy" group and lower-body power assessments. TPR = true positive rate; FPR = false positive rate; AUC = area under the curve. Arrows indicate location on curve where sensitivity is maximized and 1 specificity is minimized.

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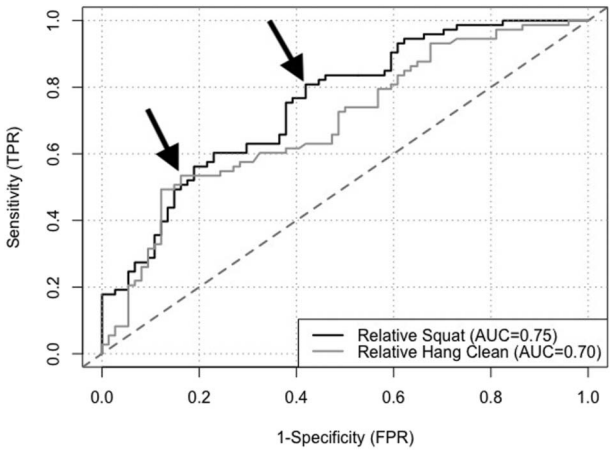

Figure 3. Receiver operator curves and AUC values for the "light" group and lower-body relative strength assessments. TPR = true positive rate; FPR = false positive rate; AUC = area under the curve. Arrows indicate location on curve where sensitivity is maximized and 1 specificity is minimized.

this model revealed an 84.6% chance of correctly classifying fast performers and an 81.0% chance of correctly classifying slower performers. Furthermore, revealing a model of similar strength (AUC = 0.848), athletes within the high body mass group were correctly classified as being in the fast group in 75.3% of cases when their relative hang clean strength levels were shown to be above a threshold of 1.28 times body mass. The high body mass group was primarily composed of athletes that were playing offensive or defensive line, as well as line backers, and heavier tight ends. Therefore, the above findings may be of acute interest to practitioners working with these position groups. Strength and conditioning practitioners may use cut-off scores (i.e., thresholds) to help inform their decision making regarding the implementation of training interventions. For instance, although speculative, and pending further longitudinal investigations, these data may suggest that improvements in maximal lower-body strength for the hang clean exercise align with improvements in linear speed until an athlete reaches a relative lower-body strength threshold of 1.28 times body mass and that further improvements in maximal strength may not align with further improvements in linear speed. Once this threshold is reached, other forms of training may be emphasized by respective practitioners, such as power development, rate of force development, or a focus on emphasizing running technique. Furthermore, the findings within this group of high body mass players align with previous findings from our group which found performance within the hang clean exercise to be a significant predictor of on-field success within collegiate American football players, playing the positions of line backer or defensive line (23).

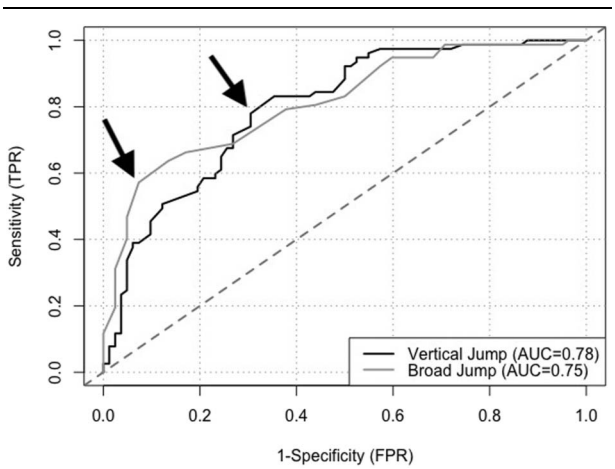

Figure 4. Receiver operator curves and AUC values for the "light" group and lower-body power assessments. TPR = true positive rate; FPR = false positive rate; AUC = area under the curve. Arrows indicate location on curve where sensitivity is maximized and 1 specificity is minimized.

Models performed better within the high body mass group of athletes suggesting that, at a simplistic level, relative lower-body strength and power thresholds seem to carry slightly more relevance in athlete groups carrying more body mass. Within the low body mass group, vertical jump height, broad jump distance, and relative squat strength showed the highest predictive ability for correctly classifying athletes into high and low linear speed groups. Interestingly, within this group of athletes, model specificity was higher for all 4 physical performance tasks, suggesting that models performed better at correctly classifying slow performers, potentially highlighting that within lighter athletes, other factors besides relative strength and power have a higher contribution on speed performance, compared with heavier athletes. Overall, the vertical jump exercises presented with the highest AUC values in both groups. This is in agreement with previous research reports showing significant correlations between vertical jump and sprint performance over varying distances and across different sport populations (12,17,26).

When interpreting the results, readers should be cognizant of some limitations pertaining the design of this study. Exploratory in nature, the investigation did not include longitudinal data, which makes it difficult to discern whether or not implementing training based on the proposed thresholds leads to substantial improvements in sprint speed. Future research should implement longitudinal research designs within similar populations to investigate the potential utility of proposed thresholds for benchmarking purposes. Specifically, when looking at the measures of strength, coaches and scientists may aim to train athletes who are below said thresholds with an emphasis on maximal strength development, and athletes who have reached said thresholds with an emphasis on other components of physical performance related to linear speed, such as power or rate of force development. If permitted by the sample size, future investigations may also investigate respective thresholds across additional subgroups (e.g., position groups). Future studies may use a similar methodological approach, looking at sprint momentum, rather than sprint velocity, as proposed in previous research reports (19). From a methodological standpoint, readers should acknowledge the limitations of measuring jump height using jump mats, which derive jump height from time spent in the air rather than take-off velocity through the impulse-momentum theorem. Jump mats have been shown to overestimate jump heights and flight times (31), and although not used in this study, correction equations may be applied where appropriate (22). Given the applied nature of the study, athlete nutrition and hydration before testing were not accounted for within the present investigation. Furthermore, although they may be viewed as a limitation of this study, the simple instrumentation used (e.g., hand timed 40-yard dash, jump mat for vertical power, and 3RM for relative strength) enhanced its ecological validity, given that these tools are available to virtually any strength and conditioning

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practitioner, and is simple to replicate. This is of importance, as it is suggested that caution should be used when trying to generalize findings beyond the sample used within a given study.

Practical Applications

This investigation puts forth different relative lower-body strength and power thresholds aimed at classifying athletes into high and low linear speed performance groups. Thresholds and model summary statistics differed based on the respective physical performance assessment and the group to which they were applied (high vs. low body mass groups). Within the high body mass group, vertical jump performance (AUC = 0.903), broad jump performance (AUC = 0.883), and relative hang clean strength (AUC = 0.848) showed the highest predictive ability regarding correctly classifying athletes into respective linear speed performance groups. These results reflect proficiency within movements involving higher rates of force development and a rapid hip extension within the group of faster athletes. Strength and conditioning practitioners may find this insightful regarding their selection of exercises enhancing such qualities. Within the low body mass group, vertical jump performance (AUC = 0.779), broad jump performance (AUC = 753), and relative squat strength (AUC = 0.74) showed the highest predictive ability regarding correctly classifying athletes into respective linear speed performance groups. In this group, the multifactorial nature of speed development was reflected within the lower AUC values and higher model specificity, which may be acknowledged by coaches training populations with a wide range of anthropometric features. Interestingly, in both groups, the vertical jump exercise showed the highest predictive ability of correctly classifying athletes into respective speed groups. Strength and conditioning practitioners may use the vertical jump as a monitoring tool when training for lower-body power to enhance sprint speed. Pending further longitudinal investigations of how increases in strength and changes in speed complement each other, methods and findings from this study may be used to benchmark athletes and to further individualize training implementations aimed at improving linear sprint speed performance through targeted interventions within the realms of strength and conditioning. Caution is advised when trying to generalize established thresholds past the population used within this study. Thresholds and findings will most likely vary based on factors such as sex, training age, sport, and measurement devices used.

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